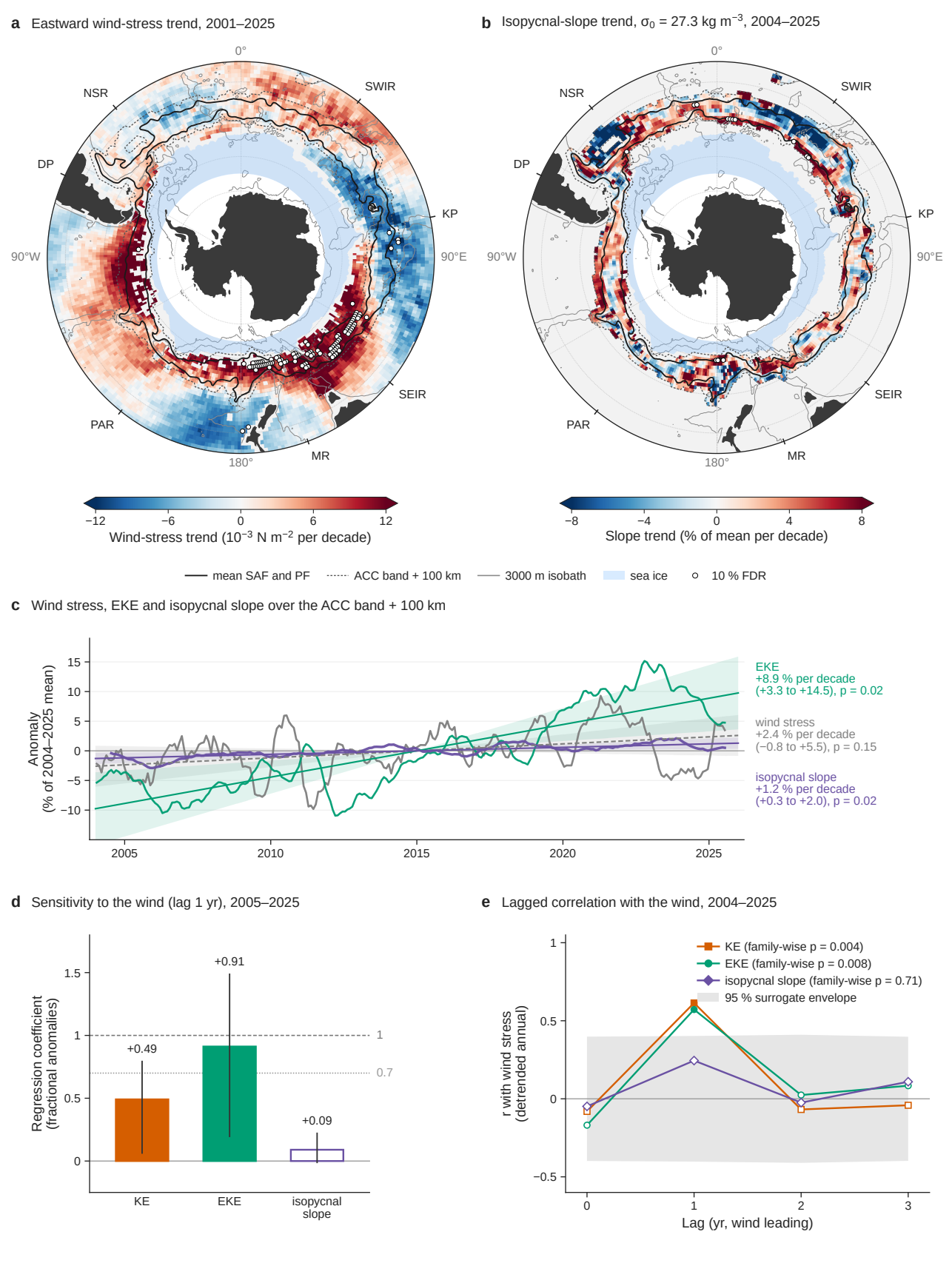

a Eastward wind-stress trend, 2001–2025
b Isopycnal-slope trend, σ0 = 27.3 kg m−3, 2004–2025
0°
NSR
SWIR
DP
KP
90°W
90°E
PAR
SEIR
MR
180°
−12 −6 0 6 12
Wind-stress trend (10−3 N m−2 per decade)
−8 −4 0 4 8
Slope trend (% of mean per decade)
mean SAF and PF
ACC band + 100 km
3000 m isobath
sea ice
10 % FDR
c Wind stress, EKE and isopycnal slope over the ACC band + 100 km
Anomaly
(% of 2004–2025 mean)
15
10
5
0
−5
−10
2005
2010
2015
2020
2025
EKE
+8.9 % per decade
(+3.3 to +14.5), p = 0.02
wind stress
+2.4 % per decade
(−0.8 to +5.5), p = 0.15
isopycnal slope
+1.2 % per decade
(+0.3 to +2.0), p = 0.02
d Sensitivity to the wind (lag 1 yr), 2005–2025
Regression coefficient
(fractional anomalies)
1.5
1
0.5
0
+0.49
+0.91
+0.09
1
0.7
KE
EKE
isopycnal
slope
e Lagged correlation with the wind, 2004–2025
r with wind stress
(detrended annual)
1
0.5
0
−0.5
KE (family-wise p = 0.004)
EKE (family-wise p = 0.008)
isopycnal slope (family-wise p = 0.71)
95 % surrogate envelope
0
1
2
3
Lag (yr, wind leading)